

JAMAL NASIR

Financial Analyst | Credit Risk & Guarantee Structuring | Quantitative Modeling & Machine Learning

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Professional Summary

Finance professional with a rare intersection of institutional credit risk expertise, guarantee structuring experience, and hands-on machine learning capabilities. I have four years of experience, one year in Pakistan's financial and public sectors and three years as an independent freelancer serving Ireland-based clients. I bridge quantitative finance, explainable AI (XAI), and full-stack analytics to solve real-world problems. Proven track record managing PKR 5.5+ billion in guarantee facilities with tier-1 banks, conducting credit evaluations across 24+ entities, and building production-grade quantitative models. Combines deep understanding of capital structure, liquidity, and coverage metrics with modern AI/ML tools (LoRA, DPO, PyTorch) to drive data-informed financial decisions.

Professional Experience

Financial Planning Analyst

May 2026 – Present

National Credit Guarantee Company Limited (NCGCL)

- Built and deployed an **export intelligence application** automating manual analytics across all Pakistani export chapters; reduced multi-day reporting workflows to a single-click dashboard. [\[Link to live app\]](#)
- Directed the LUMS-MAF returnable guarantee facility for Pakistan's swappable battery program, managing fund allocation and financial frameworks for MAF-backed capital.
- Executed transaction evaluations and issued offer letters and guarantees, including:
 - PKR 1.5 billion** facility with HBL Microfinance Bank
 - PKR 4.0 billion** facility with Bank Alfalah
- Leading Phase 2 of the Guarantee Management Portal (GMP) development, standardizing data fields and reporting protocols across all onboarded Financial Institutions (FIs).

Analyst

October 2025 – April 2026

The Pakistan Credit Rating Agency (PACRA)

- Managed credit evaluations for a **24-entity portfolio**, primarily in the energy and construction sectors.
- Analyzed capital structure, liquidity, coverage, and cash-flow metrics to assess credit risk and assign ratings.

Research Associate

August 2025 – January 2026

Project Indigenous Mechanization, Government of Punjab

- Collaborated with Dr. Shakeel Sadiq Jajja on a government initiative focused on indigenous e-mechanisation.
- Led field research engaging **50+ stakeholders** (farmers, manufacturers, banks) to map mechanisation adoption barriers and opportunities.

Executive Assistant – Strategic Planning Cell

July 2024 – December 2024

Institute of Business Administration (IBA), Karachi

- Co-developed IBA's **5-year strategic plan** across seven pillars, designing accreditation-aligned KPIs.
- Supported data modeling and dashboarding for institutional performance tracking.

Project Consultant (Remote)

October 2022 – January 2024

QuantOra, Dublin

- Built differential-equation models for Irish cattle-land saturation forecasting.
- Mentored the team on an MNIST classification pipeline using machine learning frameworks.

Key Projects

Tiny Llama Fine-Tuning (LoRA & DPO)

Spring 2025

Text Analytics

- Fine-tuned a 1.1B parameter TinyLlama model using **LoRA** and **Direct Preference Optimization (DPO)**.
- Improved instruction-following and question-answering alignment on custom evaluation datasets.

Optimization for High-Dimensional Logistic Regression

Spring 2025

- Performed extensive literature review of existing optimization techniques and mathematical derivations.
- Built and ran multiple optimization models on high-dimensional datasets.

Loan Servicing Risk Alert System

Fall 2024

Financial Data Analytics

- Designed early-warning models to flag high-risk microfinance clients.
- Identified top predictive features and supported workflow design for model deployment.

Predictive Models: Classification & Regression

Fall 2024

Machine Learning

- Built classification and regression models for two Kaggle tasks under tight deadlines.
- Used feature engineering and iterative tuning to reach competitive leaderboard performance.

Mathematics of Derivative Pricing

Summer School, IBA

Summer 2023

- Studied mathematical frameworks for futures and options pricing.
- Implemented **Black–Scholes** and **Heston models** in Python.

Option Pricing Models

Summer School, IBA

Summer 2023

- Applied **Monte Carlo simulations** on National Stock Exchange (NSE) market data.
- Benchmark-tested simulation outputs against Black–Scholes predictions.

Blockchain Fundamentals & Ethereum DApp Deployment

Summer 2022

- Set up Ethereum VM on **Microsoft Azure** and developed Solidity-based smart contracts.
- Built and tested a decentralized application (DApp) using **Remix IDE**.

Education

MS Financial Management

Lahore University of Management Sciences

2025

BS Economics & Mathematics

Institute of Business Administration (IBA), Karachi

2021

Skills

Programming & Data:	Python (NumPy, Pandas, SciPy, scikit-learn, TensorFlow, PyTorch), R, SQL
Tools & Platforms:	Microsoft Azure, Power BI, Excel Power Tools, EViews, MATLAB, STATA, SPSS, Git, LaTeX
Methods & Frameworks:	Econometrics, Forecasting, Machine Learning, LoRA, DPO, Credit Risk Analysis, Valuations and Financial Analysis

Certifications

- **IELTS** – Band 6.5
- **Blockchain Basics & Smart Contracts**